

MOHAMMED FARHAAN PATEL

SENIOR PLATFORM ENGINEER • SITE RELIABILITY ENGINEER • AI INFRASTRUCTURE ENGINEER

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PROFESSIONAL SUMMARY

Senior Platform Engineer with **7+ years** building the internal platforms a **3,300-engineer** organization ships on, powering streaming services reaching **120M+** active accounts - multi-region Kubernetes across AWS and GCP, artifact and software supply-chain infrastructure, cloud cost engineering and AI agents in production operations. Led a company-wide migration across 58 engineering teams that was scoped for two years and delivered in one quarter, earning an Engineering Spotlight award; architected a partner-facing artifact distribution platform; and put AI agents into production incident response and Kubernetes diagnostics.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript/TypeScript, Go, SQL, Bash Scripting, Groovy
- **CI/CD & GitOps:** GitLab CI/CD, GitHub Actions, Jenkins, ArgoCD, FluxCD, Spinnaker
- **Infrastructure as Code (IaC):** Terraform, Terragrunt, Ansible, CloudFormation, Helm Charts
- **Frameworks & Tools:** FastAPI, Flask, Django, React, Next.js, Streamlit, Celery, REST APIs
- **Databases:** PostgreSQL, MongoDB, DynamoDB, MySQL, Redis, Neo4j, Amazon Redshift, ClickHouse
- **Containers & Orchestration:** Kubernetes, Docker, Helm, Istio Service Mesh, Cilium, ArgoCD, Kustomize, ApplicationSets
- **Architecture:** Microservices, Event-Driven Architecture, Platform Eng, Distributed Systems, High Availability, Multi Region & Cloud
- **Cloud Platforms:** AWS (EKS, EC2, S3, Lambda, Athena, IAM, SQS, SNS, Route53, CloudWatch), GCP (GKE, GCS, Cloud Monitoring)
- **Observability & Monitoring:** Prometheus, Thanos, Loki, Grafana, Mimir, Elasticsearch, OpenTelemetry, Datadog, PagerDuty, OpenCost
- **Security & Compliance:** ModSecurity/OWASP CRS, ESO, SOPS, Trivy, SBOM, Xray, Entra ID/ Azure AD, OAuth2 Proxy, OIDC, RBAC, Zero-Trust Architecture
- **AI & Automation:** Claude & OpenAI SDK, Multi-Agent Orchestration, MCP, LangChain, K8sGPT, Prompt Engineering, LLM Gateway Routing, Agent Identity & OIDC

EXPERIENCE

Roku

Boston, MA

Senior DevOps & Platform Engineer - CTI Team | Kubernetes, AWS, ArgoCD, AI Agents

July 2024 - Present

- Engineering owner (DRI) for Roku's artifact platform - Artifactory, Xray, Distribution and Edge - accountable for architecture, upgrades, reliability and roadmap across the services every engineering team depends on to build and ship.
- Led a company-wide artifact platform migration - scoped as a 2-year program, delivered in one quarter - moving 58 engineering teams and 85 service accounts onto project-scoped repositories with isolated RBAC, using virtual-repository passthrough so teams switched publishing endpoints with zero downtime and no artifact copying. Recognized with the company Engineering Spotlight award.
- Replaced pipeline-driven deploys with GitOps across the platform: stood up ArgoCD, onboarded every owned service, and moved continuous delivery out of GitLab CI into declarative ApplicationSets - cutting release lead time, making rollback a git revert, and surfacing configuration drift in real time rather than at the next incident.
- Architected a partner-facing artifact distribution platform giving external silicon partners a public, isolated endpoint for firmware and container artifacts - HA Kubernetes edge, SSO-gated console, per-partner scoped tokens and release-bundle promotion from the primary cluster - and drove the design, network topology and cost model through architecture review.
- Surfaced \$200K+ in annual cloud savings by building a per-workload cost platform on OpenCost with AWS Cost and Usage Reports and spot pricing feeds through Glue and Athena, then acting on it - integrating kube-green to idle non-production workloads off-hours and moving eligible dev and QA fleets onto spot instances.
- Gave every team a bill for its own footprint, attributing storage and network spend to the owning project across three regions and two clouds and classifying each transfer as same-AZ, cross-AZ or cross-region so teams could see - and cut - the cost of their own traffic.
- Turned vulnerability response from a multi-team investigation into a single query by modelling the full supply chain in Neo4j and ClickHouse - developer commit, GitLab build, runner, Artifactory publish, Xray scan, cluster, namespace - so any CVE traces to who published it, what built it, and every place it runs.
- Cut mean time to resolution 40% by building TicketOps, a Claude multi-agent system that triages incoming tickets, dispatches Kubernetes, observability and code agents, and opens merge requests with proposed fixes for human review.
- Put root-cause analysis in front of engineers where they already work, building the findings API, per-environment ApplicationSets and scoped analyzer RBAC that route cluster failures through an internal LLM gateway and render plain-language diagnoses beside the owning application in ArgoCD.
- Eliminated configuration drift and cut provisioning time 80% by architecting Configuration as Code in Terraform and Python, managing repositories, permissions, LDAP group mappings and retention policies declaratively across every environment.

DevOps Engineer – CTI Team | Kubernetes, Docker, GitLab CI/CD, Python, Helm

Jun 2021 – Jun 2024

- Enabled 100+ teams to self-provision CI/CD runners in under 5 minutes by building GitLab Runners as a Service across AWS and GCP regions, with cost attribution and observability built in.
- Removed long-lived runner credentials from infrastructure code by migrating CI runner fleets from Terraform to ArgoCD GitOps, sourcing tokens at runtime from AWS Secrets Manager via External Secrets Operator so rotation became a secret update instead of a redeploy across 15+ multi-cluster environments.
- Reduced alert noise 60% and incident response time 35% with an alerting-as-code framework on ArgoCD Notifications and Terraform, routing PagerDuty, Slack and email with team-based RBAC.
- Ended stale-rotation pages for 20+ engineering teams by building a PagerDuty-to-Slack sync service that keeps each team's Slack handle resolved to whoever is actually on call.
- Cut CI/CD pipeline debugging time 50% with MeshTrace, a distributed tracing tool that follows a CI job across the service mesh, ingress and artifact backends.

- Removed a recurring class of pipeline outages by building a token lifecycle platform (FastAPI, React, PostgreSQL) that discovers every personal, group and project access token, alerts owners ahead of expiry, and surfaces stale credentials for revocation.

Perfios Mumbai, IN
Software Engineer II - Rest API and Cloud Team | Python, AWS, Bash, Docker, MongoDB May 2017 – Aug 2019

- Scaled identity verification to 250M+ API requests/day with an event-driven pipeline on AWS Lambda, S3, SQS and EMR for real-time background checks and fraud detection.
- Expanded coverage to 150+ government data sources used by 50+ enterprise financial clients by building REST APIs for identity verification and KYC compliance.
- Improved deployment frequency 10x and cut scaling response from hours to minutes by containerizing a monolith into Docker microservices.
- Cut database queries 70% and improved API latency 3x with a Redis cluster caching layer; monitored 1.2K+ API calls/sec via an ELK platform.

INTERNSHIPS

Informatica Redwood City, CA, USA
CLOUD DEVOPS INTERN - R&D AISight Team | Kubernetes, CI/CD, Chaos Engineering Oct 2020 - May 2021

- Identified 12+ failure modes before production impact by building an automated chaos engineering framework using Gremlin for scheduled resilience and reliability testing of Kubernetes workloads.
- Cut cluster node count 30% and lowered compute spend through Kubernetes pod optimization scripts for dynamic resource patching and right-sizing across deployments.

Global Angel Fund Inc. San José, CA, USA
FULL STACK INTERN - Backend and DevOps Team | Python, Javascript, Bash, Kafka, CI/CD May 2020 - Aug 2020

- Delivered the Angel Fund MVP to production 2 weeks ahead of the 3-month deadline by architecting and building the complete backend stack from scratch to deployment.
- Cut deployment time from 2+ hours to 15 minutes across 4 application stacks with fully automated CI/CD pipelines and real-time analytics dashboards.

KEY PROJECTS

TicketOps – AI Multi-Agent Infra Remediation Agent Python, Claude SDK, Kubernetes, MCP, GitLab API, Jira

Multi-agent system that triages incoming infrastructure tickets and monitors cluster health via Prometheus, dispatching specialized Kubernetes diagnostics, observability and code-remediation agents. Detects pod crashes, resource exhaustion and failed deployments, then opens merge requests with proposed fixes or executes remediation behind human-in-the-loop approval. Cut mean time to resolution 40%.

Software Supply Chain Knowledge Graph Neo4j, ClickHouse, Kubernetes Collectors, Gitlab, Artifactory, Xray, Trivy

Models the full software supply chain in Neo4j and ClickHouse - developer commit, GitLab build, runner, Artifactory publish, Xray scan, cluster and namespace - so a CVE or vulnerable package can be traced to who published it, what built it, and every place it runs.

VibeDeploy – Platform-as-a-Service FastAPI, React, MongoDB, Kubernetes

Self-service deployment platform for one-click provisioning of databases, web servers, MCP servers, AI apps and cron jobs on Kubernetes, with JWT/OAuth authentication, resource validation and cost calculation.

EDUCATION

San José State University San José, CA, USA
M.S. Computer Engineering | Specialization in Data Science | GPA: 3.5 Aug 2019 - May 2021

Pune Vidyarthi Griha's, University of Pune Nashik, IN
B.S. Computer Engineering | Minors in Cloud Computing | GPA: 3.8 May 2012 - May 2017

AWARDS & HONORS

Engineering Spotlight - company recognition for delivering a 2-year artifact platform migration program in a single quarter Roku 2026
Invited panel speaker on AI & DevOps, addressing an audience of 3,000 students K. K. Wagh College of Engineering, Nashik 2017
Graduated top of class, B.S. Computer Engineering University of Pune 2017
Elected Vice President & Secretary, Computer Engineering University of Pune 2017

PUBLICATION

Filed a patent on "[Robust Human Detection & Tracking using Human Model based Camera calibration](#)" with IJARIIIE